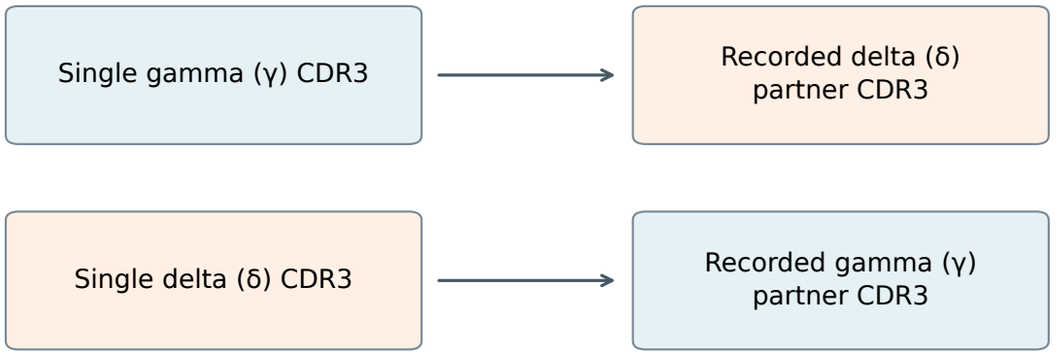
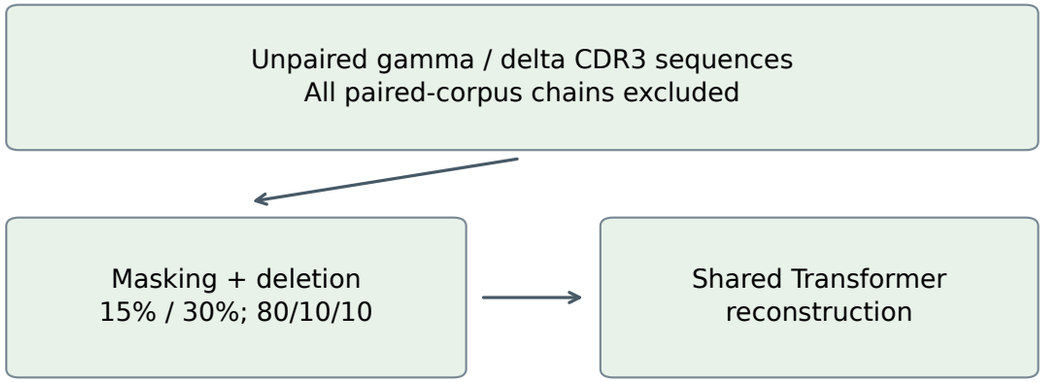
A Task definition



Two directions share one generator
Multiple recorded partners per query are retained

Endpoint: recorded sequence recovery

B Single-chain denoising pretraining

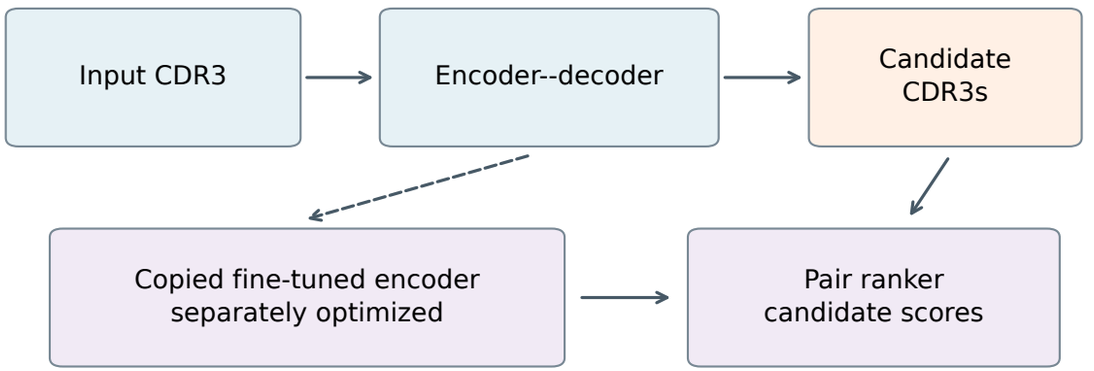


Teacher-forced token loss and aligned-copy control
Unique-sequence-disjoint train / validation

Reconstruction is a separate endpoint

C Conditional generation and separate ranking

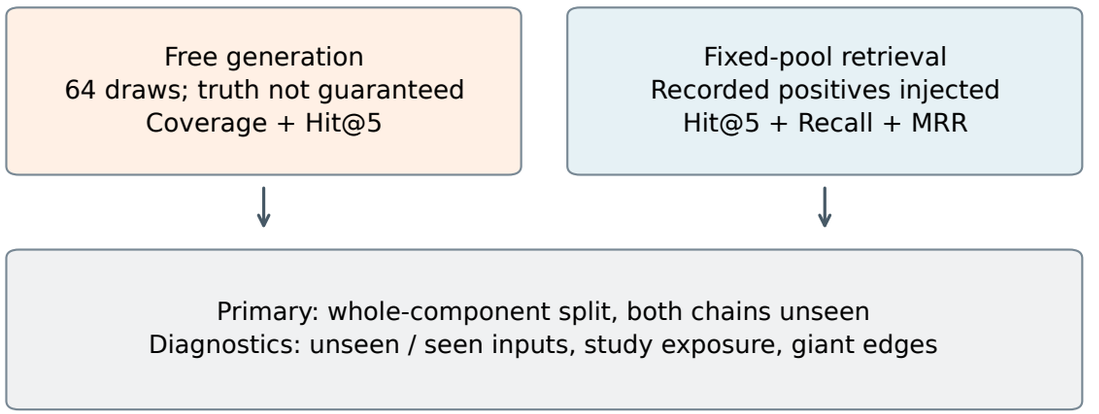
Paired fine-tuning on recorded training edges



Same training edges; original generator stays fixed

Generator: 7.40M parameters | 4 encoder + 4 decoder layers
Width 256 | 8 heads | FFN 1024 | dropout 0.1 | max 64 tokens

D Controlled evaluation



Primary: equal-direction mean query-macro Hit@5 at 64 draws

Exact-overlap and sequence-distance audits accompany recovery